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RESEARCH ARTICLE

Zoono-TransmiRisk: A Framework to Enhance Zoonotic Disease Monitoring With SHAP-Driven Insights

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ABSTRACT Zoonotic diseases, transmitted between animals and humans, pose a persistent global health challenge. Most prior studies have focused on detecting individual pathogens (e.g., influenza, brucellosis) while giving limited attention to transmission risk—the likelihood and pathways of disease spread—thus restricting their generalizability. This study is guided by the hypothesis that effective monitoring of zoonotic diseases requires distinguishing between mild, self-limiting cases and high-transmission scenarios, and that integrating a structured rule-based scoring framework with machine learning models and SHAP interpretability can systematically classify transmission risks across multiple zoonotic diseases. The dataset is structured at the disease level (one row = one disease) to enable systematic risk classification, while future extensions will incorporate case-level data for enhanced predictive validity. The framework evaluates transmission risk by assigning weighted values to epidemiological and environmental features, including transmission type, host animals, contagious period, and severity. To validate this approach, three machine learning models—Random Forest, XGBoost, and Logistic Regression—are applied. Random Forest achieves the highest accuracy (0.92), while SHAP (SHapley Additive exPlanations) analysis enhances interpretability by identifying animal involvement, transmission type, and high-risk groups as key drivers of risk. Unlike prior disease-specific approaches, the resulting Streamlit-based prototype enables real-time transmission risk evaluation under user-defined conditions (e.g., transmission mode, animal density, high-risk groups, environmental settings). By providing actionable insights and tailored recommendations—such as allocating greater resources for high-risk scenarios—the framework offers a practical decision-support tool for proactive zoonotic disease surveillance and public health response.

INDEX TERMS Zoonotic diseases, transmission risk classification, machine learning, SHAP interpretability, one health, predictive framework, disease spread potential.

I. INTRODUCTION

Zoonotic diseases, or zoonoses, are infectious diseases that are naturally transmitted from animals to humans. They are caused by diverse groups of pathogens, including bacteria, viruses, parasites, and fungi, and represent a significant global public health concern. Zoonoses account for a substantial proportion of emerging infectious diseases, with notable examples including rabies, avian influenza, Ebola virus disease, brucellosis, and COVID-19.

Transmission of zoonotic diseases can occur through direct contact with infected animals, ingestion of food or water contaminated with pathogens, bites from infected vectors (e.g., mosquitoes or ticks), or inhalation of airborne infectious agents.

The rising incidence of zoonotic outbreaks is driven by multiple interconnected factors, such as deforestation, climate change, intensification of agriculture, and increasing human–wildlife interactions. Furthermore, urbanization and

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globalization have facilitated the wider dissemination of these diseases through human mobility, international trade in animals, and movement of animal products.

In this context, the One Health approach, which recognizes the intrinsic interconnectedness of human, animal, and environmental health, has emerged as a critical framework for mitigating the risks and impacts of zoonotic diseases [1].

The integration of surveillance systems across human, animal, and environmental sectors enables early outbreak detection, improved risk assessment, and the implementation of effective control measures. Figure 1 illustrates the interconnected pathways of zoonotic disease transmission, highlighting how pathogens circulate among animals, the environment, and humans.

Wildlife frequently acts as a natural reservoir, facilitating spillover events either directly or through intermediate hosts such as livestock. Livestock populations often serve as bridge hosts, interacting with both wildlife and humans and thereby amplifying transmission risks. Domestic animals also contribute to zoonotic transmission due to their close association with humans, transmitting pathogens through bites, scratches, or contact with contaminated surfaces.

The environment constitutes an additional critical component, functioning as a reservoir where pathogens can persist in water, soil, or air. A comprehensive understanding of these transmission pathways is essential for designing integrated surveillance and control strategies, ultimately reducing outbreak risks and strengthening global public health preparedness.

The effective management of zoonotic diseases requires a comprehensive, multi-sectoral approach that integrates epidemiological surveillance, predictive modelling, and risk assessment [2], [3]. Recent technological advancements, particularly in artificial intelligence (AI) and machine learning (ML), have significantly improved the capacity to predict and monitor zoonotic disease transmission. These predictive models enable the identification of high-risk factors, characterization of transmission pathways, and provide evidence-based insights to guide targeted interventions and policymaking.

Strengthening public health awareness, implementing robust biosecurity measures, and advancing research in vaccines and therapeutics remain critical pillars of zoonotic disease prevention and control [4]. A comprehensive understanding of the multifactorial dynamics driving zoonotic transmission is fundamental to strengthening global health security. In this context, the development of integrated surveillance systems, the expansion of diagnostic capacities, and the promotion of sustained international collaboration represent essential strategies to mitigate the risk of future pandemics.

Ultimately, the integration of transdisciplinary research with evidence-based interventions provides a strategic foundation for addressing the persistent challenges posed by zoonotic diseases, thereby enhancing public health resilience in an increasingly interconnected world.

The rising incidence of zoonotic diseases is driven by multiple interrelated factors. Climate change alters the ecology and life cycles of reservoirs and vectors, thereby reshaping the geographical distribution of disease. Rapid urbanization further amplifies spillover risks by increasing human exposure to animal populations [5]. Moreover, close human–animal interactions arising from livestock production, pet ownership, and wildlife trade create diverse transmission pathways, substantially elevating the probability of zoonotic spillover. These dynamics underscore the critical need for proactive vigilance and pre-emptive risk assessment to mitigate emerging public health threats.

Recent advances in explainable artificial intelligence (XAI) have highlighted the importance of transparency and interpretability in predictive modelling. Hybrid interpretable machine learning approaches, for example, have been developed to enhance classification performance while ensuring explainability, thereby producing outputs that are both accurate and trustworthy [6]. This principle is particularly relevant for zoonotic disease risk prediction, where interpretability is essential for guiding public health interventions. While prior studies demonstrate the methodological versatility of interpretable ML, their applications have largely remained domain-specific. In contrast, zoonotic disease surveillance remains underexplored in this context, despite the urgent demand for transparent, interpretable, and scalable frameworks for transmission risk assessment.

This study addresses this gap by proposing the TransmiRisk framework, a hybrid risk assessment approach that integrates rule-based scoring with machine learning and SHAP-based interpretability to systematically classify zoonotic disease transmission risks across multiple pathogens.

Traditional zoonotic disease surveillance systems predominantly rely on retrospective data analysis, which delays outbreak detection and response. These systems are further constrained by underreporting, fragmented data sources, and the absence of real-time monitoring [7]. As a result, they are primarily reactive, identifying outbreaks only after they occur. Such a reactive approach is inadequate in the context of emerging zoonoses, where rapid detection and early intervention are critical to preventing widespread transmission.

To address these challenges, predictive modelling using machine learning has emerged as a valuable approach for assessing and predicting zoonotic disease risks [8]. Machine learning models are well-suited for processing high-dimensional, complex datasets and detecting non-linear relationships among environmental, demographic, and epidemiological factors that influence disease transmission. Leveraging historical data, these models can predict outbreak risks, thereby strengthening early warning systems and supporting targeted public health interventions.

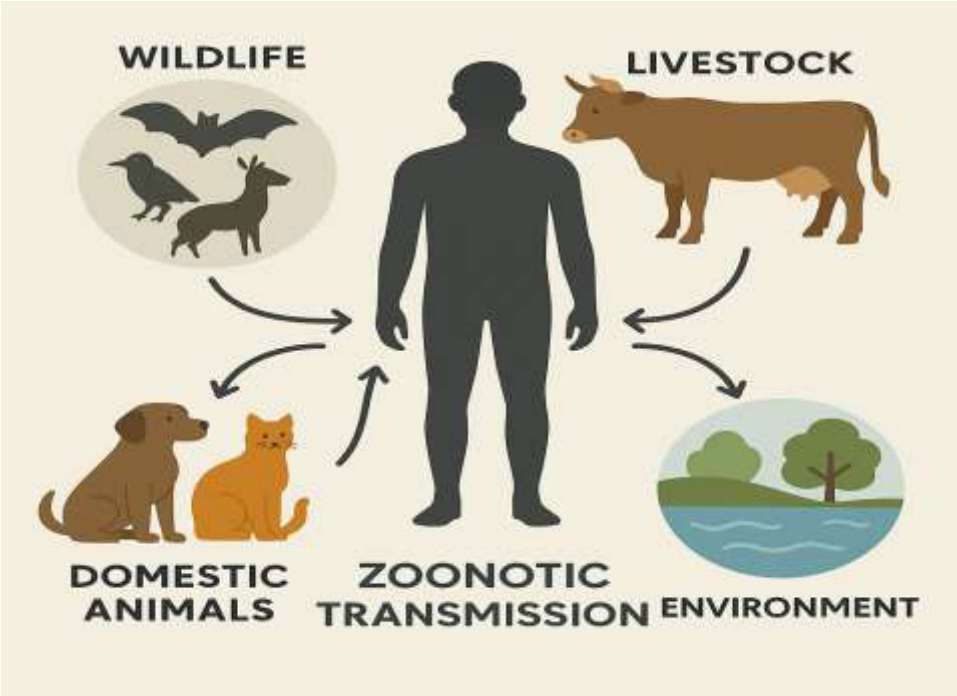


FIGURE 1. Zoonotic disease transmission pathways.

This paper introduces the Zoono-TransmiRisk Framework, a comprehensive risk assessment model designed to predict zoonotic disease transmission scenarios. The framework integrates multiple machine learning algorithms, including XGBoost, Random Forest, and Logistic Regression, to estimate transmission risks based on diverse features such as animal population dynamics, human activity patterns, and environmental conditions.

To ensure accessibility and practical utility, the framework is deployed through Streamlit, offering an interactive, real-time platform for risk evaluation. This deployment enables public health practitioners and policymakers to monitor, predict, and respond to zoonotic disease risks more effectively, ultimately enhancing preparedness and response capacity. Accordingly, this study is guided by two central research questions: (1) Can a hybrid framework that integrates rule-based scoring, machine learning, and SHAP interpretability reliably classify zoonotic diseases into Low, Moderate, and High-risk levels? and (2) Which epidemiological factors exert the strongest influence on these risk predictions? These hypotheses frame the methodological design and evaluation presented in this work.

Unlike prior studies that primarily focus on detecting the presence of zoonotic diseases or maximizing predictive accuracy, this work introduces the TransmiRisk framework, which systematically classifies transmission risk—the likelihood that a zoonotic pathogen may spread from animals to humans under specific epidemiological and environmental conditions. Transmission risk in this context reflects how factors such as transmission type, animal involvement, contagious period, and host vulnerability collectively shape the probability of disease spread. The framework integrates a rule-based scoring system, machine learning classifiers, and SHAP-based interpretability into a unified approach, ensuring both predictive capability and transparency. In addition, we present a Streamlit-based prototype tool that demonstrates practical usability, enabling researchers and public health officials to generate real-time risk evaluations and tailored prevention strategies. To our knowledge, this is the first study to combine risk scoring, predictive modeling, interpretability, and tool deployment for zoonotic disease transmission risk assessment within a One Health perspective.

The key contributions of the proposed work establish a novel risk assessment strategy for zoonotic diseases. The study provides the following main contributions:

- Development of a generalized risk assessment framework for systematically classifying zoonotic disease transmission risk, guided by a structured rule-based scoring system and validated using machine learning models (Random Forest, XGBoost, and Logistic Regression).
- Design of a Risk Scoring Framework that integrates epidemiological and environmental dimensions—including transmission type, animal reservoirs, high-risk human groups, contagious period, and severity—into categorical risk levels.
- Integration of SHAP interpretability to identify and rank the most influential features driving transmission risk, thereby enhancing transparency, interpretability, and trust in machine learning predictions.

- Deployment of the Zoono-TransmiRisk tool as a Streamlit-based interactive web interface, enabling real-time risk prediction and visualization with tailored prevention recommendations for public health users.

The rest of the paper is structured as follows: Section II provides a review of related works. Section III describes the methodology used for developing the Zoono-TransmiRisk Framework, including data preparation, risk scoring, model development, and system implementation. Section IV presents the experimental results, model evaluations, and SHAP-based interpretability analysis. Section V discusses future work directions, such as real-time data integration and system expansion to strengthen proactive zoonotic disease surveillance, as well as the limitations of this study. Finally, Section VI provides the conclusion, highlighting the findings and the practical implications for zoonotic disease risk assessment.

II. LITERATURE REVIEW

Zoonotic diseases influence global healthcare to a substantial extent because they create 60% of new pathogens that affect humans and result in one billion infections annually producing millions of fatalities [9], [10]. The healthcare burden is heavy on the economy since it damages trade relations and visitors' confidence and service availability specifically among underserved populations who lack medical care access [10]. Control and prevention strategies embrace the One Health approach that combines human health with animal health and environmental health sectors to support data collaboration and data exchange [11]. Multisectoral collaborations should focus on the effective development and deployment of vaccines since human and animal vaccination remains underutilized [12]. Preventing the spread of zoonoses requires the active management and regulation of human activities that facilitate disease transmission from animals to humans, as an integral component of strengthening public health outcomes [9]. Effective monitoring and response to zoonotic diseases require strengthened surveillance systems in conjunction with sustained international cooperation [13].

The interface between wildlife, domestic animals, and human populations creates conditions for zoonotic disease outbreaks, shaped by a combination of ecological factors and socio-economic determinants. Transmission can occur through direct contact, foodborne contamination, and vector-borne pathways, while drivers such as invasive species, habitat destruction, and climate change further amplify the spread of disease [14], [15]. The risk of zoonotic diseases intensifies when human encroachment into wildlife habitats is coupled with expanding international trade, creating opportunities for the emergence of novel pathogens [10]. The urgent need for effective public health surveillance underscores the importance of integrated approaches that link veterinary medicine with human healthcare, particularly given that zoonotic diseases account for nearly 60% of newly emerging infectious diseases [10].

Zoonotic disease transmission is strongly influenced by animal hosts and vectors, which act as intermediaries bridging the spread of pathogens between wildlife and humans. Among vector-borne pathogens, mosquitoes and ticks are particularly significant, as their blood-feeding activities transmit viruses, bacteria, and parasites responsible for substantial global morbidity and mortality [16], [17]. Behavioural modifications in vectors induced by viral infections can enhance transmission efficiency, as infected insects often exhibit altered host-seeking preferences, favouring specific vulnerable species [18]. Controlling vectors that transmit zoonotic diseases necessitates comprehensive management strategies, with Integrated Vector Management (IVM) representing a particularly effective approach. IVM combines environmental, chemical, and biological measures to disrupt vector transmission cycles, even in challenging settings where resistance to conventional insecticide-based methods is prevalent [19]. Effective vector management for disease prevention must account for environmental drivers such as climate change and urbanization, as integrating these factors helps optimize control strategies [17], [20].

Assessing disease risk is critical for both individual healthcare management and population-level health planning. Effective risk assessment enables strategic healthcare delivery, strengthens preventive measures, and supports more efficient resource allocation. However, reliance on clinical indicators alone remains insufficient for accurately determining cardiovascular disease risk at the individual level [21]. Researchers require advanced systems to assess thrombosis and bleeding risks, as comprehensive evaluation must encompass all components of Virchow's triad: hypercoagulability, endothelial dysfunction, and abnormal blood flow within vessels [21]. Farm-based biosecurity measures tailored to specific operational characteristics help agricultural systems reduce the risk of disease introduction [22]. Risk assessment models play a pivotal role in epidemic prediction and prevention. Approaches such as Markov chain analysis and the Analytic Hierarchy Process (AHP) have been applied to direct prevention efforts toward specific targets [23]. Personalized risk assessments represent an essential component of disease management; however, they remain costly and difficult to implement consistently due to resource limitations in certain healthcare environments. The main challenge lies in achieving precision while ensuring effective and scalable implementation.

The effectiveness of risk assessment directly influences healthcare resource allocation, enabling targeted medical strategies that maximize efficiency and improve clinical outcomes. Risk stratification methods are widely employed by healthcare providers to classify patients according to their likelihood of adverse outcomes, thereby guiding treatment decisions and facility resource management. For example, patients with pulmonary arterial hypertension (PAH) at high risk require intensive healthcare support, including prolonged hospital stays and emergency care [24]. Predictive modelling further enhances resource optimization by forecasting disease

patterns, detecting vulnerable patient groups, and enabling efficient distribution of healthcare services [25]. Risk assessments also strengthen treatment decisions, as demonstrated in Crohn's disease management, where patients adjust their behaviour and treatment strategies according to their assessed risk level [26]. Evidence suggests that low-risk patients often experience a higher quality of life than high-risk patients, highlighting the critical role of risk assessment in improving patient care [24]. Nevertheless, misinterpretation of risk factors or misclassification of patients can result in overtreatment and unnecessary consumption of healthcare resources.

Zoonotic diseases, which account for approximately 60% of newly emerging human pathogens—of which 71% originate from wildlife—require rigorous risk analysis [27]. Effective risk assessments allow veterinary professionals to detect and prioritize pathogens according to severity, despite global resource constraints in veterinary services [28]. Rising human exposure to zoonotic pathogens, driven by biodiversity loss and ecological change, further necessitates comprehensive ecological assessments [29]. Collaboration between veterinary and medical experts enhances the detection of transmission patterns and facilitates improved prevention strategies [30]. The SpillOver framework, for instance, has been applied to assess the spillover potential of viruses, thereby informing public health interventions and setting research priorities [31].

Risk analysis remains essential in mitigating human health threats from zoonoses, particularly in light of pandemics caused by animal-borne pathogens. Studies evaluating the zoonotic potential of SARS-related coronaviruses emphasize the importance of such frameworks for strengthening future preparedness [32]. Similarly, the management of canine tularemia highlights the necessity of coordinated strategies between human health authorities and veterinary professionals in preventing transmission among at-risk populations [33]. Risk assessment serves as an essential measure to decrease the threats that zoonotic diseases present.

The advancement of explainable artificial intelligence (XAI) has further strengthened the role of predictive modelling in healthcare. SHapley Additive exPlanations (SHAP) analysis enhances transparency by quantifying feature contributions and revealing critical interactions within complex datasets. SHAP provides intuitive single-graph visualizations that facilitate the interpretation of biological and medical data, improving hypothesis generation and decision-making [34]. For instance, in liver disease diagnosis, SHAP has demonstrated its ability to identify intricate relationships among clinical features while simultaneously improving predictive accuracy [35]. Similar benefits have been observed in diabetes risk prediction, where SHAP revealed the combined importance of General Health Status and Body Mass Index [36]. SHAP has also proven effective in hyperspectral image classification, offering per-patch explanations that clarify model predictions in high-dimensional datasets [37]. Overall, SHAP plays a critical role in advancing responsible AI by making complex machine learning models transparent and interpretable [38].

Recent studies have extended SHAP applications to zoonotic disease contexts by integrating machine learning with network analysis, thereby enabling assessment of animal reservoirs and predictor values relevant to future outbreak risks [39], [40]. As a feature attribution methodology, SHAP has demonstrated both accuracy and interpretability across domains, with notable success in heart failure prediction, achieving 91.39% accuracy [41]. However, computational demands remain a limitation, leading to the development of more efficient variants such as clustered-SHAP (C-SHAP), which improves processing speed while maintaining prediction accuracy [42]. Furthermore, integration with large language models (LLMs) enhances SHAP's usability, providing clearer explanations for non-expert users [43]. Ongoing methodological improvements continue to strengthen SHAP's role as a powerful tool for transparent, interpretable, and scalable risk assessment across diverse domains, including zoonotic disease prediction [44]. Recent studies in educational data mining have demonstrated the value of SHAP-based interpretability in predictive modelling. For example, [45] applied an LSTM-SHAP approach for academic performance prediction, while subsequent research extended this work by integrating SHAP with Bi-LSTM architectures to enhance interpretability in deep learning models. Although these applications were developed within the educational domain, they underscore the broader methodological relevance of combining machine learning with SHAP explanations—a direction that this study extends to zoonotic disease risk assessment.

III. ZOONO-TRANSMIRISK FRAMEWORK DESIGN

The overall workflow for developing the TransmiRisk Framework is illustrated in Figure.2. The outlined workflow establishes a standardized procedure to evaluate zoonotic diseases through accurate prediction methods featuring interactive functionality. The workflow functions as a whole to unite different phases of development starting from data setup through tool deployment which creates an efficient and scalable pathway for public health officials.

A. DATA PREPARATION

As illustrated in Figure. 2, Data Preparation forms the first stage of the TransmiRisk framework. This stage ensures the dataset is structured for reliable model training through preprocessing, feature engineering, and handling of missing values. While not all zoonotic diseases have severe impacts—some cause only mild, self-limiting symptoms such as fever or cold that resolve within a few days—effective monitoring requires identifying the conditions under which specific diseases are likely to spread rapidly. Addressing this distinction forms the central purpose of this study, which aims to systematically classify transmission risk across zoonotic diseases to support proactive surveillance and targeted

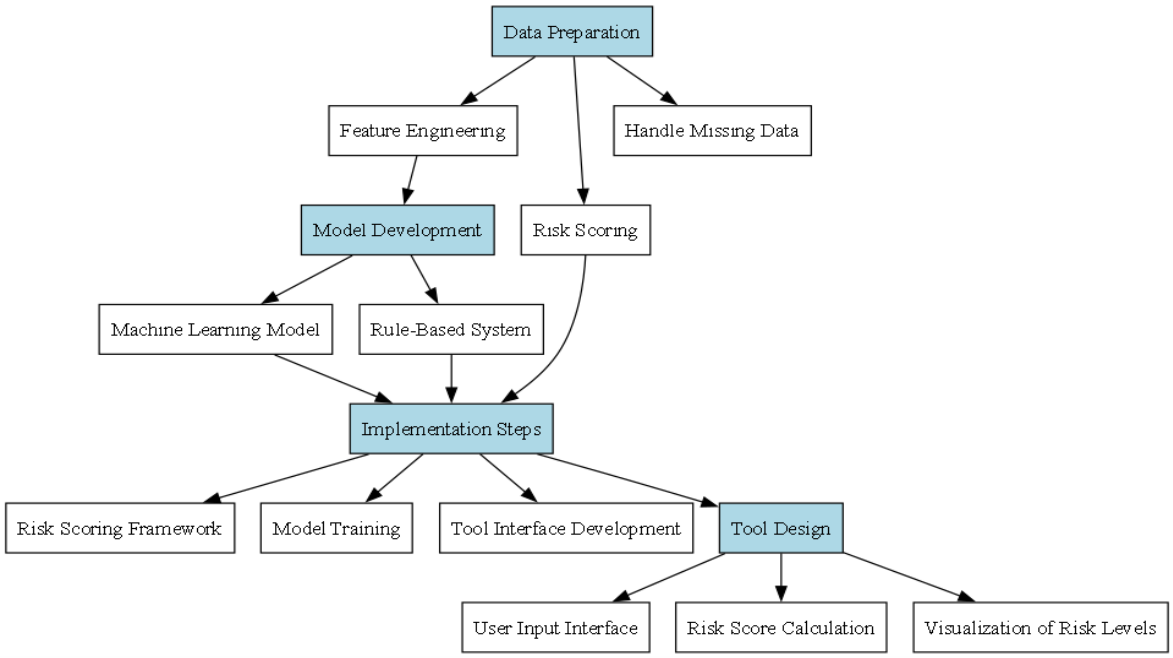


FIGURE 2. Zoono-TransmiRisk framework.

interventions. To enable this, a comprehensive and structured dataset was required. A novel dataset was created using publicly available information from reputable sources such as the World Health Organization (WHO), the Centers for Disease Control and Prevention (CDC) One Health resources, and peer-reviewed scientific literature. The dataset compiled for this study consists of 63 zoonotic diseases, each represented by 21 epidemiological features. The dataset is designed to capture the complete story of each zoonotic disease, including how it emerges, how it spreads, who it affects, and how long it remains a threat, thereby ensuring a holistic foundation for transmission risk assessment. Diseases span across multiple categories, including livestock-associated (e.g., brucellosis, anthrax), wildlife-associated (e.g., rabies, Ebola), companion-animal–related (e.g., cat scratch disease), and avian diseases (e.g., avian influenza), ensuring coverage of the major transmission pathways. Data were compiled from global published sources, providing a broad geographical representation; however, we acknowledge that some regions, particularly low-income and resource-limited settings, are underrepresented. The feature set was designed to reflect One Health dimensions, incorporating attributes from the animal domain (hosts, reservoirs, animals involved), the human domain (affected populations, age groups, high-risk groups), and the environmental domain (common environments, transmission contexts). This structure enables integration of cross-sectoral information within the risk assessment framework, while also highlighting the need for future expansion to improve geographic and socioeconomic coverage.

Inclusion criteria: Diseases were selected based on their zoonotic nature, documented human-to-animal transmission, and availability of epidemiological information across 21 predefined features (e.g., transmission type, animals involved, contagious period, risk groups).

Exclusion criteria: Diseases with insufficient or missing data across critical variables, or those not classified as zoonotic in WHO/CDC reports, were excluded to maintain dataset reliability thereby ensuring privacy by design. The compiled dataset and accompanying preprocessing code will be made publicly available upon acceptance of this manuscript to facilitate reproducibility.

1) DATASET DESCRIPTION

The dataset includes detailed information on various zoonotic diseases, covering aspects such as their causative pathogens, transmission modes, affected organs, diagnostic methods, treatments, and prevention strategies. In our dataset, environmental and socioeconomic determinants were explicitly captured through the features Common Environments (e.g., farms, urban areas, wildlife settings) and High-Risk Groups (e.g., occupational exposure, demographic vulnerabilities). These variables were integrated with clinical, pathogen-related, and transmission attributes to provide a holistic basis for risk modeling. The dataset was structured to reflect a One Health perspective by incorporating human-related attributes (Symptoms, Affected Age Group, High-Risk Groups), animal-related attributes (Animals Involved, Transmission Method), and environmental attributes (Common Environments), ensuring a multi-sectoral representation of zoonotic disease transmission. The dataset was constructed entirely from secondary sources at a categorical level (e.g., disease, pathogen type, animals involved, environments) and did not contain any personally identifiable human or animal data. No patient-level or animal-level identifiers are included

in the dataset; therefore, no anonymization procedures were required. To illustrate the dataset structure, a partial snapshot with five representative rows is provided. The dataset is divided into two structured tables. Table 1 - Disease Overview and Table 2 - Clinical and Public Health Data. The Table 1 allows for a comparative understanding of how different zoonotic diseases spread and affect humans, providing essential information for disease classification. The Table 2 helps researchers understand disease clinical effects along with public health management methods for zoonotic diseases.

2) DATA PREPROCESSING

To improve data quality and ensure robust model performance, several preprocessing steps were applied. Records with incomplete or inconsistent entries in critical variables were excluded. Missing values in categorical features were imputed using the mode, while numerical variables were imputed using the median. Categorical variables, including transmission type, animal categories, and risk groups, were transformed through one-hot encoding. Continuous variables such as contagious period were normalized using min–max scaling. To reduce redundancy and multicollinearity, correlated features were reviewed, and a final set of 21 features was retained. The dataset was divided into training (70%) and testing (30%) subsets for model development and evaluation. The distribution of class labels in the dataset was examined to assess imbalance. The initial rule-based classification produced three classes: Low (N = 58), Moderate (N = 4), and High (N = 1). Because the High class contained only one entry, it was merged into the Moderate class for machine learning experiments, resulting in two target categories: Low and Moderate. While some imbalance was present, we did not apply resampling techniques (e.g., SMOTE), as these introduce synthetic data that could bias predictions in this sensitive context. Instead, model performance was evaluated with metrics such as precision, recall, and F1-score, which are more informative under imbalance than accuracy alone. This decision preserved data authenticity while ensuring fair model evaluation. Descriptive statistics of the dataset were compiled to summarize the distribution of features across 63 zoonotic diseases. The dataset consisted of categorical variables (e.g., *Disease, Pathogen Type, Transmission Type, Animals Involved, Organs Involved, Transmission Method, Common Environments, Diagnostic Test, Prevention Methods, High-Risk Groups*), numerical variables (e.g., *Duration, Typical Recovery Time, Contagious Period*), and binary variables (e.g., *Effective Antibiotics, Additional Treatments*). Missing values were addressed through imputation, with mode replacement applied to categorical features and mean replacement applied to numerical features. Categorical variables were label-encoded to ensure model compatibility, while numerical features were normalized to maintain comparability across scales. These preprocessing steps standardized the dataset and ensured its suitability for machine learning model training.

3) FEATURE ENGINEERING

Effective feature engineering is critical for the development of predictive models, as it improves both accuracy and interpretability. The process began with feature selection, in which only statistically significant variables with strong influence on prediction outcomes were retained. This step reduced noise by eliminating irrelevant or redundant information.

Subsequently, feature transformation was applied to ensure the dataset was in a suitable format for analysis. Techniques such as normalization, standardization, and categorical encoding were employed to maintain consistency across diverse attributes, thereby enhancing comparability and improving model performance.

Finally, feature generation was implemented to create more informative variables by combining existing data sources. For instance, integrating animal contact frequency with environmental conditions and human activity levels enabled the construction of risk indices that provided deeper epidemiological insights. In addition, a Risk Level column (Low, Moderate, High) was derived using the risk scoring framework and incorporated into the dataset as the target variable for model training. The detailed scoring methodology and framework are described in the Implementation section.

4) HANDLING MISSING DATA

Missing values in epidemiological datasets can undermine the reliability of predictive models if not appropriately addressed. To mitigate this, different strategies were applied depending on the type and extent of missing data. For numerical features, simple statistical methods such as mean or median imputation were employed, while categorical features were imputed using the mode (most frequent value). In cases where only a small proportion of records contained missing values, deletion was considered to minimize bias. When simple imputation proved insufficient, missing values were estimated using published epidemiological studies or historical disease trends.

Simple imputation approaches were selected because they preserve dataset interpretability, introduce minimal computational overhead, and are less prone to overfitting compared to more complex methods such as KNN or multiple imputation. This balance between practicality and robustness ensured that the dataset remained complete, consistent, and suitable for zoonotic disease risk prediction.

B. MODEL DEVELOPMENT

The second stage, Model Development (Figure. 2), builds both the rule-based system and the machine learning models. This stage establishes the foundations for risk classification by combining rule-based scoring with predictive algorithms. The methodological design directly tests the hypothesis that integrating rule-based risk scoring with machine learning classifiers improves predictive performance while preserving interpretability. While deep learning approaches such as Graph Neural Networks (GNNs) and Long Short-Term

TABLE 1. Disease overview (Partial view of the dataset prepared (subset of 5 rows)).

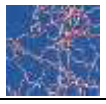
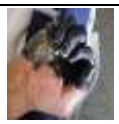
Disease	Pathogen			Transmission Type	Animals Involved	Affected Human Organs	Transmission Method	Symptoms	
	Type	Picture	Name					Common Symptoms	Untreated Skin Symptoms
Rabies	virus		Rabies lyssavirus	Direct contact	Dogs, bats, raccoons, foxes	Central nervous system	Animal bites, saliva contact	Fever, headache, excess salivation, muscle spasms, paralysis	
Lyme Disease	bacterium		Borrelia burgdorferi	Vector-borne	Deer, mice, ticks	Skin, joints, heart, nervous system	Tick bites	Fever, chills, headache, fatigue, muscle and joint aches, Bulls eye rash	
Plague	bacterium		Yersinia pestis	Vector-borne	Rodents, fleas	Lymph nodes, lungs, blood	Flea bites, contact with infected animals	Fever, headache, chills, weakness, swollen lymph nodes	
West Nile Virus	virus		West Nile virus (WNV)	Vector-borne	Birds, mosquitoes	Central nervous system	Mosquito bites	Fever, headache, body aches, joint pains	
Hantavirus	virus		Puumala hantavirus	Air-borne	Rodents	Lungs	Inhalation of rodent droppings, urine, saliva	Fever, muscle aches, shortness of breath	

TABLE 2. Clinical and public health data (Partial view of the dataset prepared (subset of 5 rows)).

Disease	Description	Affected Age Group	Common Environments	Duration	Diagnostic Test	Effective Antibiotics	Additional Treatments	Typical Recovery Time	Prevention Methods	Contagious Period	High-Risk Groups
Rabies	Severe neurological disease caused by Lyssavirus	All ages	Areas with high animal contact	Days to weeks	DFA test on brain tissue, saliva, serum or skin biopsy	None	Supportive care, vaccine post-exposure	Varies based on exposure to treatment	Vaccination of animals, avoiding wild animals	Varies from days to weeks	People with frequent animal exposure
Lyme Disease	Multisystem inflammatory disease transmitted by ticks	All ages, particularly outdoor active individuals	Wooded and grassy areas	Weeks to months	ELISA test, Western Blot	Doxycycline, Amoxicillin, Cefuroxime	Pain relievers, supportive care	Weeks to months	Avoiding tick-infested areas, using tick repellents	Weeks to months	Outdoor workers, hikers
Plague	Severe infectious disease caused by Yersinia pestis	All ages	Areas with high rodent populations	Days to weeks	Blood test, lymph node aspiration, chest X-ray	Streptomycin, Gentamicin, Doxycycline	Hospitalization, supportive care	Weeks to months	Reducing rodent habitats, using flea control measures	Days to weeks	People in areas with poor sanitation
West Nile Virus	Mosquito-borne viral infection	All ages, especially older adults	Urban and rural areas	Weeks to months	Blood test, lumbar puncture	None	Pain relievers, supportive care	Weeks to months	Avoiding mosquito bites, using mosquito repellents	Weeks to months	Outdoor workers, older adults
Hantavirus	Severe respiratory disease caused by hantavirus	All ages	Rural areas, buildings with rodent infestations	Days to weeks	Blood test, chest X-ray	None	Intensive care, mechanical ventilation	Weeks to months	Reducing rodent exposure, proper cleaning methods	Weeks to months	People in rural or forested areas

Memory (LSTM) networks have shown promise in zoonotic prediction tasks, they were not included in this study due to the relatively small, disease-level dataset (63 samples across 21 features). Applying such models under these conditions would risk overfitting and reduce the validity of the results. Developing a predictive model for zoonotic disease

risk assessment requires a structured methodology that combines data-driven learning with manually assigned scores derived from key epidemiological factors. The objective of model development is to enhance accuracy in identifying high-risk scenarios while ensuring interpretability to support decision-making. Epidemiological and clinical data were used to construct predictive frameworks capable of classifying disease risks. Two complementary approaches were employed: a machine learning model, which applies statistical algorithms to detect patterns and forecast risk levels, and a rule-based system, which assigns risk scores based on manually defined epidemiological attributes. Integrating these approaches provides a balance between automation and expert-driven assessment, thereby strengthening surveillance capacity and enabling earlier detection of zoonotic threats.

The TransmiRisk framework was implemented in Python and deployed through a Streamlit-based interface, allowing interactive exploration of risk predictions. At present, the framework is being evaluated in a research setting and has not yet been deployed in live healthcare systems.

1) MACHINE LEARNING MODEL

Supervised machine learning models were initially explored to classify zoonotic diseases into different transmission risk categories. The models were trained on the constructed dataset using selected epidemiological and clinical features. Specifically, three algorithms were employed: Logistic Regression (LR), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). Logistic Regression was selected for its simplicity, interpretability, and ability to generate probabilistic outputs for classification tasks. Random Forest was chosen due to its robustness, capability to handle non-linear relationships, and effectiveness in reducing overfitting through ensemble learning. XGBoost was included as a state-of-the-art boosting algorithm that offers superior performance in capturing complex feature interactions and improving predictive accuracy. However, despite their complementary strengths, these models were unable to independently differentiate between low, moderate, and high risk, as the dataset did not contain explicit indicators mapping directly to these categories. This limitation highlighted the challenge of applying purely data-driven techniques in the absence of sufficient labelled data. Consequently, while machine learning provided a foundation for identifying patterns within the dataset, the final classification of risk levels required the integration of a rule-based system with manually defined scoring rules to ensure reliability and interpretability.

2) RULE BASED SYSTEM

The rule-based scoring framework was introduced to provide a transparent and interpretable foundation for transmission risk classification. By encoding established epidemiological principles (e.g., transmission type, host animals, contagious period, symptom severity, and high-risk groups), the framework ensures that risk levels are grounded in domain knowledge rather than solely statistical patterns. This structured baseline not only supports decision-making for public health officials but also serves as a foundation for validating and complementing machine learning models.

The rule-based system operates on manually assigned scores derived from key epidemiological factors. In this study, the system applies thresholds from the risk scoring framework to classify diseases into categorical risk levels. The thresholds were set using a logical scaling approach to distinguish between low, moderate, and high transmission risks:

- **Low Risk**: Total Score < 10
- **Moderate Risk**: $10 \leq$ Total Score < 15
- **High Risk**: Total Score ≥ 15

This scoring-based rule system was also used to generate the Risk Level column (Low, Moderate, High) in the dataset, which served as the target variable during model training. The approach ensures that classification remains transparent and interpretable, while also providing a baseline against which machine learning predictions can be compared.

Together, the rule-based system and machine learning models form the core of the proposed predictive framework. The rule-based system provides a transparent baseline for classification, while the machine learning models capture complex patterns to enhance predictive capability. Having established these complementary approaches, the next step involves detailing their operational implementation, including how the risk scoring framework is applied, how the models are trained and validated, and how the system is deployed through an interactive interface for practical use in zoonotic disease surveillance.

C. IMPLEMENTATION STEPS

The third stage, Implementation Steps (Figure. 2), integrates the rule-based scoring framework, machine learning model training, and tool interface development. This stage operationalizes the workflow by linking risk scoring with automated predictions and decision-support features. The implementation phase operationalizes the developed framework by translating the designed models and scoring logic into a practical system for zoonotic disease risk assessment. This stage ensures that predictive methods are systematically applied and effectively integrated to support public health decision-making. The process consists of four major components: (i) the Risk Scoring Framework, which standardizes disease attributes into categorical risk levels; (ii) the Risk Characterization and Workflow, which illustrates the factors and decision flow underpinning classification; (iii) Model Training and Validation, which establishes the predictive capacity of machine learning algorithms; and (iv) Tool Development, which integrates the framework into an interactive interface with risk computation, visualization, and deployment capabilities. Together, these steps ensure that the framework is not only scientifically robust but also practically applicable for real-time surveillance. The rule-based scoring framework first generates categorical risk levels (Low, Moderate, High). These labels serve as the dependent variable

for machine learning model training, ensuring consistency between rule-based classification and data-driven prediction. The ML models learn from the input features and attempt to replicate the rule-based classification, thereby providing an automated predictive counterpart. This hybrid structure allows the framework to reconcile rule-based scores with machine learning outputs, integrating transparency and predictive flexibility.

1) RISK SCORING FRAMEWORK

The risk scoring framework was designed to systematically classify zoonotic diseases into categorical risk levels based on key epidemiological factors. This framework assigns scores to attributes such as transmission type, animals involved, contagious period, symptom severity, and high-risk groups, with higher scores reflecting greater transmission potential and public health impact. The cumulative risk score for each disease is then mapped to predefined thresholds, enabling its classification into Low, Moderate, or High risk categories. By translating heterogeneous characteristics into standardized numerical values, the framework ensures a transparent, consistent, and reproducible method for disease risk assessment. The scoring framework is detailed in Table 3.

TABLE 3. Risk scoring criteria for zoonotic disease.

Category	Factor	Score
Transmission Type	Airborne	5
	Vector-borne	4
	Direct contact	3
	Waterborne	3
	Foodborne	2
	Other	1
Animals Involved	Pets	4
	Livestock	3
	Wildlife	2
	Rodents	5
High-Risk Groups	Veterinarians	5
	Outdoor workers	4
	General population	3
Contagious Period	Days	1
	Weeks	3
	Months	5
Severity of Symptoms	Severe symptoms (e.g., organ damage)	5
	Moderate symptoms	3
	Mild symptoms	2

Each risk factor—transmission type, animals involved, high-risk groups, contagious period, and symptom severity—was scored on a 1–5 ordinal scale, where 5 denotes very high risk and 1 represents minimal risk. The scoring was derived from well-established patterns of zoonotic disease spread. For example, airborne transmission was assigned the highest score (5) due to its association with large-scale outbreaks such as influenza and COVID-19, while foodborne transmission was given a lower score (2) to reflect its comparatively limited spread. Similarly, rodents were scored as 5 because of their recurrent role in high-risk zoonoses such as plague and hantavirus, whereas wildlife reservoirs were assigned a lower score (2). High-risk occupational groups such as veterinarians and immunocompromised individuals were scored 5, outdoor workers 4, and the general population 3, reflecting differences in exposure. Longer contagious periods (months) and severe clinical outcomes were also weighted higher (score = 5) compared to short-duration illnesses and mild symptoms (score = 1–2).

D. CLASSIFICATION BASED ON TOTAL SCORE

To categorize zoonotic disease risk, a cumulative scoring approach is used, where the total score is derived by summing individual factor scores from the risk scoring framework. The Total Risk Score for each disease was computed as the sum of factor scores is given in the equation (1)

$$\text{Total Score} = \sum_{i=1}^{n} \text{Factor Score}_i \tag{1}$$

The classification criteria are outlined in Table 4. This classification provides a structured method for evaluating the severity of zoonotic disease threats. A High-Risk designation ($\geq$15) indicates a significant public health concern requiring immediate intervention. A Moderate Risk score ($>$10) suggests a disease with potential for controlled spread, necessitating continuous monitoring. A Low-Risk classification ($<$10) represents a minimal threat but still requires preventive measures to mitigate transmission. This framework supports a systematic approach to prioritizing disease management strategies.

TABLE 4. Risk level classification for zoonotic diseases.

Total Score	Risk Level
≥ 15	High Risk
> 10	Moderate Risk
< 10	Low Risk

E. RISK CHARACTERIZATION FRAMEWORK

The risk characterization framework provides a structured representation of the key factors influencing zoonotic disease transmission and severity. It emphasizes five primary dimensions: transmission type, animals involved, contagious period, symptom severity, and high-risk groups, each capturing critical epidemiological attributes that determine how diseases emerge and spread within populations. The numerical weights assigned to features (e.g., Airborne = 5, Vector-borne = 4) were not arbitrarily chosen but derived from relative risk rankings consistently reported in WHO, CDC, and published epidemiological literature. These qualitative distinctions were mapped onto a standardized 1–5

scale to enable systematic scoring and comparison across diseases. The framework is illustrated in Figure. 3, where these dimensions and their subcategories are organized to provide a clear basis for systematic risk assessment.

(1) Transmission type – reflects the primary mode of spread, such as airborne, direct contact, or vector-borne pathways.

(2) Animals involved – considers the role of pets, livestock, and wildlife as reservoirs or carriers.

(3) Contagious period – accounts for the duration of infectiousness, ranging from days to weeks or months.

(4) Symptom severity – highlights the clinical impact, distinguishing between mild, severe, and fatal outcomes.

(5) High-risk groups – identifies vulnerable populations, including children, the elderly, and immunocompromised individuals.

By integrating these dimensions, the framework enables a holistic view of zoonotic disease risk. It not only supports the development of scoring and classification methods but also enhances interpretability by linking quantitative risk assessment to epidemiological reasoning.

F. ZOONOTIC DISEASE RISK FLOWCHART

The Figure 4 illustrates the workflow of the zoonotic disease risk assessment system. The zoonotic disease risk flowchart detailing the sequential process by which diseases are evaluated and classified into risk categories. It integrates both the risk characterization framework and the risk scoring framework, thereby linking qualitative attributes with quantitative assessment. Beginning with the identification of epidemiological factors—such as transmission type, animals involved, contagious period, symptom severity, and high-risk groups—the flowchart demonstrates how these inputs are systematically scored, aggregated, and compared against predefined thresholds. This stepwise process enables the transparent classification of diseases into Low, Moderate, or High risk levels. By visually mapping the flow from attribute assessment to final classification, the flowchart provides an intuitive understanding of the framework's operation. It also serves as a bridge between data preparation and implementation, ensuring clarity in how structured epidemiological information translates into actionable risk categories.

1) MODEL TRAINING

The machine learning models were trained on the prepared zoonotic disease dataset, where the Risk Level column (Low, Moderate, High) served as the target variable. Three supervised classification algorithms—Random Forest, XGBoost, and Logistic Regression—were employed to capture different patterns in the data.

Prior to training, the dataset was preprocessed through normalization, encoding of categorical variables, and handling of missing values to ensure compatibility with model requirements. Hyperparameter tuning was performed using grid search combined with k-fold cross-validation to identify optimal configurations and reduce the risk of overfitting.

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive view of each model's ability to correctly classify diseases into their respective risk categories. The comparative results highlight the relative strengths of the algorithms, thereby supporting the selection of the most effective predictive approach for zoonotic disease risk assessment. To evaluate model performance, we applied stratified 5-fold cross- validation to preserve the class distribution across folds, with repeated runs conducted to confirm the stability of results. Furthermore, SHAP (SHapley Additive Explanations) analysis was incorporated to enhance interpretability, allowing the contribution of individual features to model predictions to be systematically assessed.

G. XGBOOST (EXTREME GRADIENT BOOSTING)

Extreme Gradient Boosting (XGBoost) is an optimized ensemble learning algorithm based on decision trees, widely recognized for its efficiency and predictive accuracy in classification tasks. The algorithm builds multiple weak learners (decision trees) sequentially, where each new tree corrects the residual errors of the previous trees. This iterative process is guided by the principle of gradient boosting, which minimizes the loss function through gradient descent optimization.

A key strength of XGBoost lies in its ability to handle high-dimensional data while reducing overfitting through techniques such as regularization (L1 and L2 penalties), shrinkage (learning rate adjustment), and subsampling of rows and columns. These mechanisms enhance both robustness and generalizability, making the algorithm particularly suitable for risk classification in heterogeneous datasets.

In this study, XGBoost was trained using the prepared zoonotic disease dataset with the Risk Level column as the target variable. Hyperparameter tuning was performed to optimize the number of trees, maximum tree depth, and learning rate. Model performance was subsequently evaluated using accuracy, precision, recall, and F1-score, and the feature contributions were further analyzed using SHAP values to ensure interpretability of predictions.

Extreme Gradient Boosting (XGBoost) is an ensemble learning algorithm that constructs additive decision trees to minimize a specified loss function through gradient boosting. At each iteration, a new weak learner is added to correct the errors of the previous model, which can be represented in Equation (2)

$$F_{t(x)} = F_{\{t-1\}(x)} + \eta \cdot w_j \tag{2}$$

where η is the learning rate, and w_j is the weight adjustment based on gradient and Hessian values. The optimization process relies on the use of first- and second-order derivatives of the loss function. Specifically, the gradient (g_i) and Hessian (h_i), for each instance i are computed by Equations (3) and (4)

$$g_t = \left(\partial L\left(y_t, \hat{y}_t\right)\right) / \left(\partial \hat{y}_t\right) \tag{3}$$

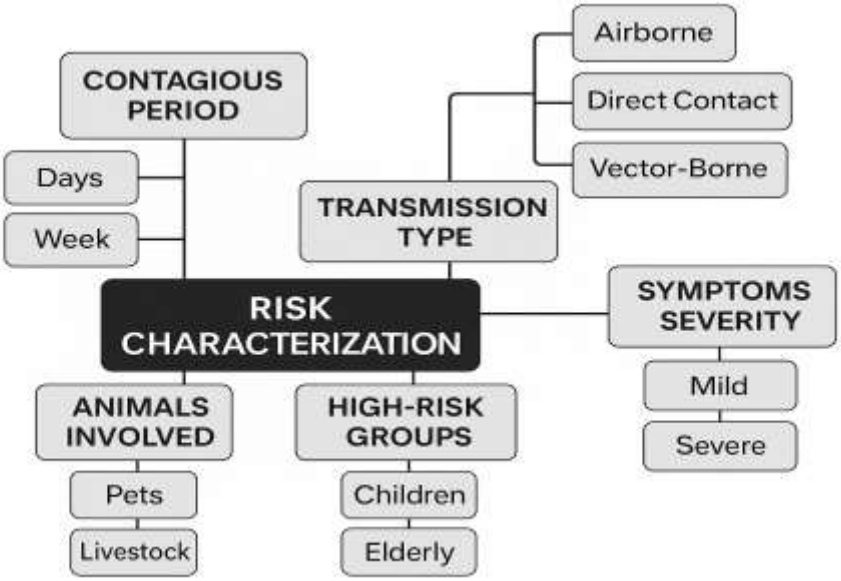


FIGURE 3. Risk characterization framework.

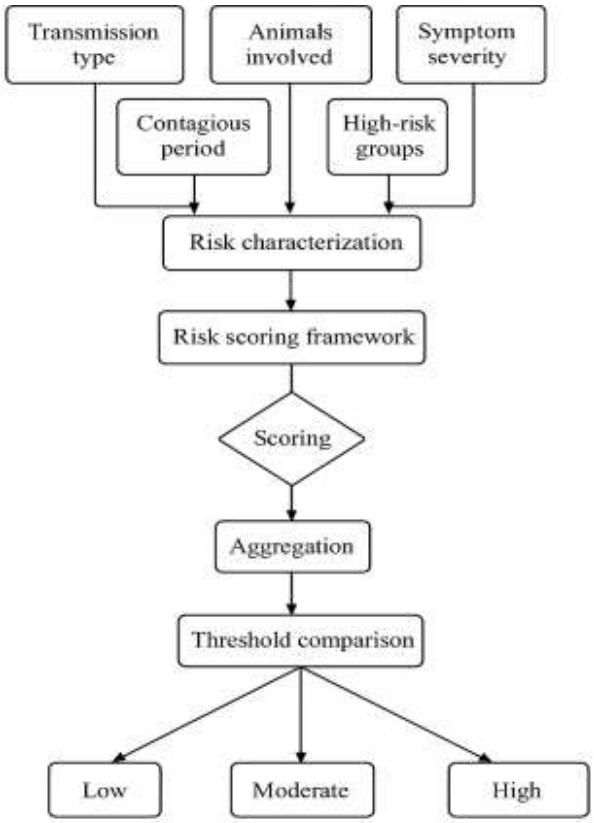


FIGURE 4. Zoonotic disease risk flowchart.

$$h_i = \left(\partial^2 L\left(y_i, \hat{y}_i\right)\right) / \left(\partial \hat{y}_i^2\right) \tag{4}$$

For binary classification tasks, such as distinguishing between high-risk and low-risk categories, the log-loss function is defined by Equation (5)

$$L = -\sum \left[y_t \log\left(\hat{y}_t\right) + (1 - y_l) \log\left(1 - \hat{y}_l\right)\right] \tag{5}$$

This formulation allows XGBoost to efficiently approximate complex non-linear decision boundaries while maintaining robustness through regularization and second-order optimization. In this study, the model was tuned using cross-validation to optimize hyperparameters such as tree depth, learning rate, and number of estimators.

H. RANDOM FOREST

Random Forest is an ensemble learning algorithm that combines the predictions of multiple decision trees to improve classification accuracy and reduce overfitting. Each tree is trained on a bootstrap sample of the dataset, and at each split, a random subset of features is considered to increase diversity among trees. Each decision tree in Random Forest is trained on a bootstrap sample, and the final prediction is obtained by majority voting across all trees in the forest is defined by Equation (6)

$$\mathrm{P(y/x)} = \frac{1}{N} \sum\nolimits_{i=1}^{N} h_i(x) \tag{6}$$

This reduces overfitting and enhances generalization. where $h_i(x)$ is the prediction of the ith tree. The splitting criterion for decision trees is Gini Impurity, given by Equation (7)

$$G = 1 - \sum\nolimits_{i=1}^{c} p_t^2 \tag{7}$$

A lower Gini value indicates a purer node, which is preferred for better decision boundaries. where p_i is the proportion of class i samples in a node c is the number of classes.

By averaging predictions over multiple trees and reducing variance through bagging, Random Forest achieves strong

generalization performance. In this study, Random Forest was applied to classify zoonotic diseases into Low, Moderate, and High risk categories, with the number of trees and maximum depth tuned using cross-validation.

I. LOGISTIC REGRESSION

Logistic Regression is a statistical classification model that estimates the probability of an instance belonging to a particular class using the logistic (sigmoid) function. Unlike linear regression, which predicts continuous values, logistic regression maps predictions to a bounded interval [1, 0], making it suitable for binary and multiclass classification tasks. Logistic Regression was included as a baseline model due to its interpretability, despite its limitation of assuming linear separability, which may oversimplify complex epidemiological interactions. The probability that an instance x_i belongs to the positive class is modeled as shown in Equation (8)

$$P\left(y_i = 1/x_i\right) = \sigma\left(w^T x_i + b\right) = 1/1 + \exp\left(\mathrm{w^T x_1 + b}\right) \tag{8}$$

where w is the weight vector, b is the bias term, and $\sigma(\cdot)$ represents the sigmoid function.

The model is trained by minimizing the log-loss (binary cross-entropy) function across all training samples as expressed in Equation (9)

$$L = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(\hat{y}_i\right) + (1 - y_i)\log\left(1 - \hat{y}_i\right)\right] \tag{9}$$

where N is the number of training instances, y_i is the true label, and $\hat{y}_i$ is the predicted probability. By adjusting the parameters w and b to minimize this loss, Logistic Regression provides a linear decision boundary for classification. In this study, Logistic Regression was employed to classify zoonotic diseases into Low, Moderate, and High risk categories, with the one-vs-rest strategy applied for multiclass extension.

Together, these three algorithms represent complementary approaches to classification. XGBoost leverages gradient boosting with advanced regularization, making it highly effective for capturing complex non-linear patterns. Random Forest provides robustness through bagging and majority voting, reducing variance and overfitting while maintaining interpretability through feature importance.

Logistic Regression, although simpler, offers transparency and a strong statistical foundation for classification, serving as a baseline against which ensemble models can be compared. By employing these diverse methods, the framework balances predictive power, robustness, and interpretability, ensuring reliable assessment of zoonotic disease risks.

J. PERFORMANCE METRICS

To evaluate the effectiveness of the classification models, multiple performance metrics were employed, ensuring a comprehensive assessment of predictive capability. The following metrics were calculated based on the confusion matrix:

(1) **Accuracy**, is defined as the proportion of correctly classified instances among the total number of samples, is given by equation (10)

$$\text{Accuracy} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{10}$$

(2) **Precision** is defined as the ratio of correctly predicted positive instances to the total predicted positives, is given by equation (11)

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{11}$$

(3) **Recall (Sensitivity)** is defined as the ratio of correctly predicted positive instances to all actual positives, is given by equation (12)

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{12}$$

(4) **F1-Score** is defined as the harmonic mean of precision and recall, which balances the trade-off between the two, is given by equation (13)

$$\begin{aligned}&\text{F1} - \text{Score}\\ &= 2 \times [(\text{Precision} \times \text{Recall})/(\text{Precision} + \text{Recall})]\end{aligned} \tag{13}$$

Here TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

These metrics together provide a robust evaluation framework. Accuracy measures overall correctness, while Precision and Recall capture performance under class imbalance. The F1-score offers a balanced view by penalizing extreme disparities between precision and recall. In this study, these metrics were applied to compare model outputs, enabling the identification of the most reliable algorithm for zoonotic disease risk classification.

1) TOOL INTERFACE DEVELOPMENT

The final step, interface design and development, ensures that the risk assessment model is interactive, user-friendly, and accessible to veterinarians, epidemiologists, and researchers. The interface enables users to input epidemiological data such as environmental factors, animal exposure, and pathogen characteristics to generate real-time risk scores. Key features include interactive dashboards, visualization of risk levels, and automated reporting to facilitate interpretation. Furthermore, cloud-based deployment supports scalability and real-time updates, enabling dynamic disease surveillance. By integrating these components, the implementation phase delivers a zoonotic disease risk assessment system that is scientifically robust, operationally efficient, and practically useful for enhancing public health preparedness and response.

K. TOOL DESIGN

The final stage, Tool Design (Figure. 2), provides the interactive environment for zoonotic risk assessment. It enables user input, risk score calculation, and visualization of classification outcomes, translating the framework into a

practical application. As part of the tool design, the TransmiRisk framework was implemented in Python with a Streamlit-based interface to provide an interactive demonstration of its functionality. The architecture integrates two core components: (i) the rule-based scoring system, which assigns categorical risk levels (Low, Medium, High); (ii) the machine learning classifiers (Random Forest, XGBoost, and Logistic Regression), which generate predictive outcomes. For demonstration purposes, the tool was applied to the compiled dataset of 63 zoonotic diseases with 21 features, showcasing its ability to classify diseases and provide interpretable results in a user-friendly environment. The design phase focuses on developing an inclusive and interactive zoonotic disease risk assessment platform that enables seamless user interaction, accurate risk estimation, and effective visualization of results. The system is expected to handle diverse epidemiological factors while maintaining efficiency and usability. Key design considerations include scalability to support evolving datasets, adaptability to incorporate emerging risk factors, and interoperability with existing surveillance infrastructures. In addition, security mechanisms must be incorporated to safeguard sensitive health data. The ultimate objective is to provide an effective and accessible decision-support system that assists researchers, healthcare professionals, and policymakers in assessing zoonotic risks and making evidence-based decisions.

1) USER INPUT INTERFACE

The user input interface serves as the primary entry point for researchers, clinicians, and policymakers to provide relevant data for zoonotic disease risk assessment. It must be intuitive, user-friendly, and capable of supporting diverse epidemiological and clinical parameters, including transmission modes, animal reservoirs, and symptom severity. The design should accommodate multiple input types, such as text fields, dropdown menus, and checkboxes, to ensure compatibility with a wide range of applications. Real-time validation mechanisms are essential to prevent errors such as omissions or inconsistent entries. An interactive interface enhances usability while ensuring accuracy and efficiency in the risk assessment process.

2) RISK SCORE CALCULATION

The risk score module is the computational engine of the tool that takes user-input data and creates a measurable risk estimate. It is either based on supervised machine learning algorithms trained on epidemiological data or rule-based systems where risk is assigned according to predetermined scoring rules. The backend system computes automatically, combining scores on multiple factors such as transmission mode of the disease, host species, and population susceptibility. For better reliability, the system can include statistical weighting procedures to rank strong contributors to the risk of zoonotic diseases. The generated score is thereafter translated into categorical risk categories (e.g., low, moderate, high), providing a normalized method of assessing risk.

3) VISUALIZATION OF RISK LEVELS

To ensure maximum interpretability, the final risk assessment is presented in interactive and visual forms. Graphical representations such as heatmaps, bar plots, and risk matrices enable users to instantly grasp the severity of a specific disease scenario. For instance, risk hotspots can be indicated through color-mapped maps, whereas the temporal dynamics of disease transmission can be represented through time-series plots. Furthermore, categorical risk labels provide a straightforward classification, enabling rapid decision-making for public health action. With the incorporation of dynamic visualization techniques, the tool not only improves access but also enables evidence-based policy formulation in zoonotic disease control.

4) TOOL DEPLOYMENT USING STREAMLIT

The TransmiRisk Framework has been implemented as an interactive web application using the open-source Streamlit library, which enables the development of lightweight, data-driven applications. The interface is designed with two primary modules:

1) **User Input Module:** This module allows users to provide key epidemiological parameters, including animal host information, environmental conditions, and transmission modes. The interface is adaptive, supporting different data entry formats, which enhances user experience, feedback efficiency, and data collection speed.
2) **Visualization Module:** In the current version, the visualization module displays the calculated risk score, the corresponding risk level (Low or Moderate), and suggested prevention strategies aligned with the risk category.At present, the SHAP-based interpretability plots are not integrated; however, future versions will incorporate SHAP visualizations to enhance transparency and explainability of model outputs.

The tool executes real-time risk scoring using pre-trained machine learning models (Random Forest, XGBoost, and Logistic Regression). Based on user input, the system generates categorical predictions of 'Low' or 'Moderate' transmission risk levels.

The TransmiRisk Framework has been implemented as an interactive web application using the open-source Streamlit library, enabling real-time user interaction with risk scoring modules. The interface allows public health professionals and researchers to input key parameters such as transmission type, animal density, high-risk groups, and environmental conditions. Based on these inputs, the tool instantly computes risk scores and assigns risk levels (Low, Moderate, or High), while also generating tailored prevention recommendations. This ensures real-time responsiveness to user-provided data.

The current version of the TransmiRisk tool is implemented in Python with a Streamlit interface, supported by a lightweight backend. The application loads the curated dataset, applies the rule-based risk scoring framework, and

executes pre-trained machine learning models (Random Forest, XGBoost, Logistic Regression) for prediction. All computations are handled locally within the Python environment, with no external database or cloud dependency in the current prototype. This design ensures portability and reproducibility for research use, though it remains a static, research-focused system at present. Although the interface processes user inputs dynamically, it does not yet incorporate automated ingestion of live epidemiological surveillance data (e.g., case-level reporting systems, environmental monitoring feeds). As such, the ''real-time'' functionality in this version refers specifically to instantaneous processing of user inputs, while integration of continuous, real-time data streams remains an aspirational feature for future iterations. Planned developments include containerization using Docker and cloud-based deployment (AWS/Google Cloud), which will enable scalability, data feed integration, and pilot deployment within health systems.

IV. RESULTS AND ANALYSIS

A. PERFORMANCE EVALUATION

The final preprocessed dataset consisted of 63 zoonotic diseases with 21 features and no missing values, providing a consistent basis for model training and evaluation. Table 5 presents the performance of the rule-based system and machine learning models using overall accuracy and macro-averaged Precision, Recall, and F1-scores. Macro-averaged metrics were emphasized to provide a more reliable comparison under class imbalance, as they weight each class equally regardless of frequency. The results show that the rule-based system achieved perfect performance (1.00 across all metrics), reflecting the deterministic nature of its scoring logic. While this establishes the internal consistency of the rule-based framework, it does not substitute for external validation, which remains a priority for future work. Among the machine learning models, Random Forest demonstrated the strongest performance with an accuracy of 0.92 and a macro F1-score of 0.82, highlighting its ability to handle feature variability and correlations effectively. XGBoost obtained moderate performance (accuracy 0.77, macro F1-score 0.63), while Logistic Regression struggled with the imbalanced dataset, producing a macro F1-score of only 0.46. The poor performance of Logistic Regression in detecting the Medium risk class reflects the constraint of its linearity assumption, which limits its ability to capture non-linear dependencies among epidemiological factors. This outcome reflects the challenges posed by class imbalance and the model's linear decision boundary. These results directly reflect the class imbalance in the dataset, where Medium and High risk categories were underrepresented. The High-risk class contained only one case in the dataset and was merged into the Moderate class for machine learning experiments; therefore, results for High-risk are not reported (indicated by ''–'' in Table 5). These results confirm that ensemble approaches such as Random Forest are better suited for this dataset, while more data would be required to fully leverage advanced models such as XGBoost.

To evaluate model performance, we applied stratified 5-fold cross-validation, ensuring that class proportions were preserved in each fold. This approach provided a balance between bias and variance trade-offs, and was considered more appropriate than leave-one-out CV given the small dataset size and the risk of unstable estimates.

Unlike prior studies that report higher predictive performance using large, case-level datasets with balanced distributions, our analysis was based on a disease-level dataset of 63 zoonotic diseases with imbalanced classes, which inherently constrained achievable performance.

1) PRECISION ANALYSIS

For the Low-risk class, all models performed well, with Random Forest and XGBoost achieving perfect precision (1.00) and Logistic Regression reaching 0.92. However, for the Moderate-risk class, Random Forest achieved 0.50, XGBoost 0.25, and Logistic Regression failed completely (0.00). This indicates that Moderate-risk cases were particularly challenging for all models.

2) RECALL ANALYSIS

Recall for the Low-risk class was high across models (Random Forest and Logistic Regression: 0.92, XGBoost: 0.75). For the Moderate-risk class, Random Forest and XGBoost achieved perfect recall (1.00), whereas Logistic Regression again failed entirely (0.00). This shows that while ensemble models were able to capture the minority class, Logistic Regression lacked robustness.

3) F1-SCORE ANALYSIS

The harmonic mean of precision and recall further demonstrates these trade-offs. For the Low-risk class, Random Forest (0.96), Logistic Regression (0.92), and XGBoost (0.86) performed strongly. For the Moderate-risk class, Random Forest achieved 0.67, XGBoost 0.40, and Logistic Regression 0.00. Only Random Forest achieved balanced performance across both classes.

4) IMPLICATIONS OF THESE FINDINGS

These findings underscore that Random Forest offers the best overall balance, but class imbalance remains a critical challenge. Moderate-risk cases were severely underrepresented in the dataset, which contributed to weak macro scores for XGBoost and Logistic Regression. Future improvements will require resampling strategies (e.g., SMOTE), weighted loss functions, or larger datasets to improve classification of underrepresented classes.

B. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was conducted to uncover relationships between variables and to better understand the underlying structure of the dataset. The analysis employed a correlation matrix, complemented by feature

TABLE 5. Performance analysis.

Model	Classes Evaluated	Accuracy	Precision (Low)	Precision (Moderate)	Precision (High)	Recall (Low)	Recall (Moderate)	Recall (High)	F1-Score (Low)	F1-Score (Moderate)	F1-Score (High)	Macro-Avg Precision	Macro-Avg Recall	Macro-Avg F1-Score
Rule-Based	3 (Low, Moderate, High)	1	1	1	1	1	1	1	1	1	1	1	1	1
Random Forest	2 (Low, Moderate)	0.92	1	0.5	–	0.92	1	–	0.96	0.67	–	0.75	0.96	0.82
XGBoost	2 (Low, Moderate)	0.77	1	0.25	–	0.75	1	–	0.86	0.4	–	0.63	0.88	0.63
Logistic Reg.	2 (Low, Moderate)	0.85	0.92	0	–	0.92	0	–	0.92	0	–	0.46	0.46	0.46

importance analysis and SHAP (SHapley Additive Explanations), to enhance interpretability of the machine learning decision process.

1) CORRELATION MATRIX ANALYSIS

The correlation matrix highlights the pairwise relationships among the encoded categorical variables in the dataset. Since the dataset consists entirely of categorical attributes, Label Encoding was applied prior to computing correlations. The resulting heatmap (Figure. 5) visually represents the strength and direction of these relationships, where values closer to 1 indicate strong positive correlation, values closer to –1 indicate strong negative correlation, and values near 0 suggest weak or no correlation.

2) OBSERVATIONS FROM THE HEATMAP

The diagonal values of the matrix equal 1.0, indicating perfect self-correlation for each variable. Beyond this, several noteworthy associations emerge:

- Disease ↔ Pathogen Name (0.53): Suggests a strong link between specific diseases and their causative pathogens.
- Duration ↔ High-Risk Groups (0.52): Indicates that disease duration may vary depending on the demographic group affected.
- Environment-Related Factors ↔ Duration (0.42): Implies a possible connection between environmental exposures and the persistence of disease.

These correlations provide important preliminary insights into how different epidemiological variables interact, serving as a foundation for model training and interpretation.

C. SHAP ANALYSIS FOR MODEL INTERPRETABILITY

The implementation of SHAP (SHapley Additive Explanations) provided interpretability by quantifying the contribution of individual features to model predictions. SHAP plots illustrate the magnitude and direction of feature effects, enabling identification of significant risk factors influencing zoonotic disease classification. For this study, SHAP analysis was applied to subsets of zoonotic diseases in order to evaluate feature impacts across different prediction cases.

The results, shown in Figure. 6, highlight that the Animals Involved feature is the most influential determinant of model predictions. Positive SHAP values ranging from +0.07 to +0.31 indicate that host species substantially increase predicted risk levels, emphasizing the importance of species-based surveillance in zoonotic disease monitoring. By contrast, the Transmission Type feature consistently demonstrated a negative contribution, with SHAP values between −0.07 and −0.30, suggesting that certain transmission pathways reduce predicted risk scores, possibly due to mitigation by public health interventions and environmental controls. The Pathogen Type feature showed minimal effect, with SHAP values close to zero (0 to −0.01), implying limited influence on model predictions compared to host species and transmission channels.

Overall, the SHAP assessment confirms that animal involvement plays a dominant role in determining disease risk, while transmission mechanisms act as moderating factors. These findings underscore the need for targeted surveillance strategies focusing on high-risk host species and call for further research into transmission pathways to improve prevention and control measures.

Figure 7 presents the SHAP analysis for the middle four zoonotic diseases, highlighting the three dominant factors influencing model predictions: Transmission Type, Pathogen Type, and Animals Involved.

The Transmission Type feature emerged as the most influential factor across these cases. Positive SHAP values, such as +0.06, indicate that certain transmission pathways substantially increased predicted risk scores. This suggests that direct, vector-borne, or environmental transmission channels strongly affect the model's classification of zoonotic disease risk.

The Pathogen Type feature displayed predominantly negative contributions, with SHAP values as low as −0.04, signifying that specific pathogen classes (e.g., bacterial, viral, parasitic, or fungal) reduce predicted risk scores. While pathogen classification is important for epidemiological research, its influence on predictive outcomes was less significant compared to transmission pathways.

The Animals Involved feature demonstrated inconsistent effects across diseases, generally exerting weak negative influences ranging from −0.01 to −0.02. This suggests that

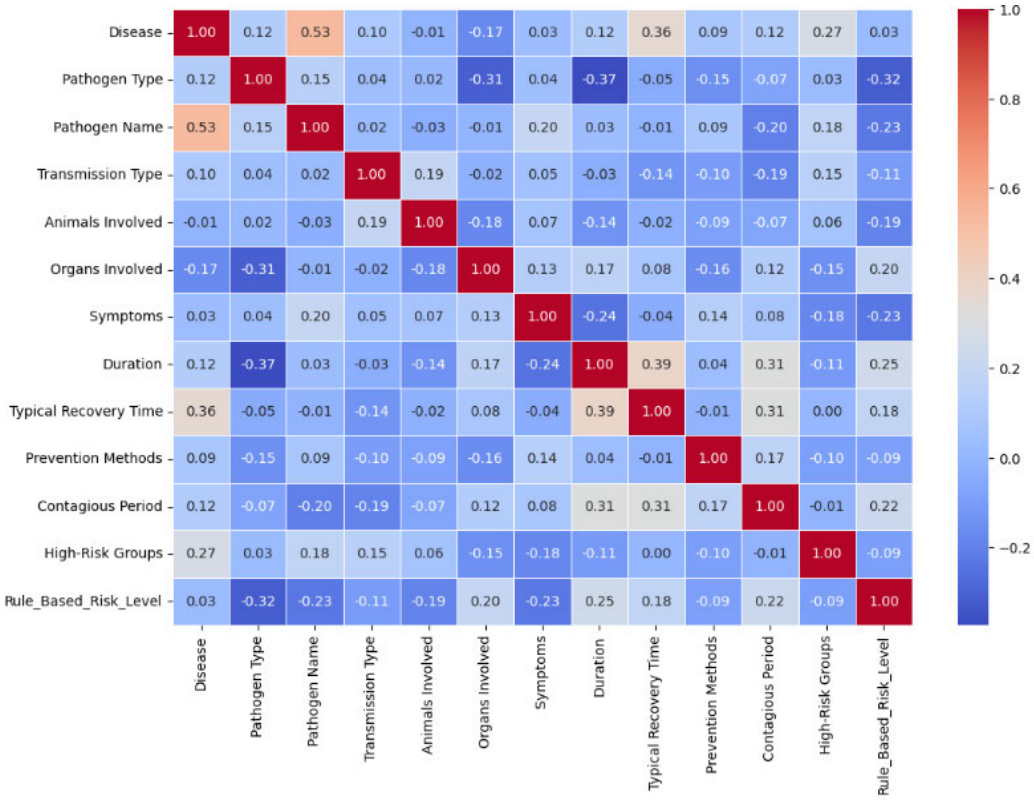


FIGURE 5. Correlation Heatmap.

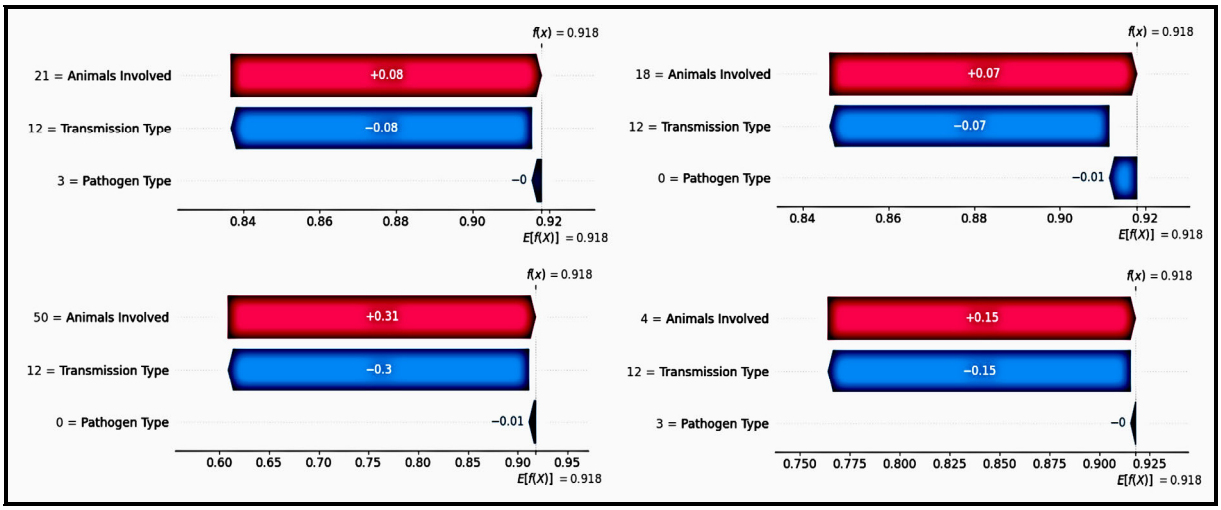


FIGURE 6. SHAP analysis of First Four Diseases (Rabies, Lyme Disease, Plague, West Nile Virus).

while host species remain relevant, their contribution to risk prediction was comparatively lower in these specific cases.

Overall, these findings highlight that transmission pathways exert the greatest influence on risk prediction, while pathogen type and animal involvement play secondary roles. For effective surveillance, strategies should integrate pathogen-specific information with host–pathogen relationships, as well as genomic and ecological data, to improve predictive accuracy. These results further underscore the importance of a One Health approach, linking veterinary and human health insights to account for species-specific transmission dynamics.

The SHAP analysis of the last four zoonotic diseases (Figure 8) highlighted Animals Involved as the dominant factor driving predicted risk scores. Its contribution was consistently positive, with SHAP values ranging from +0.02 to +0.13, reflecting the varying influence of different host species on transmission dynamics. These results reaffirm that

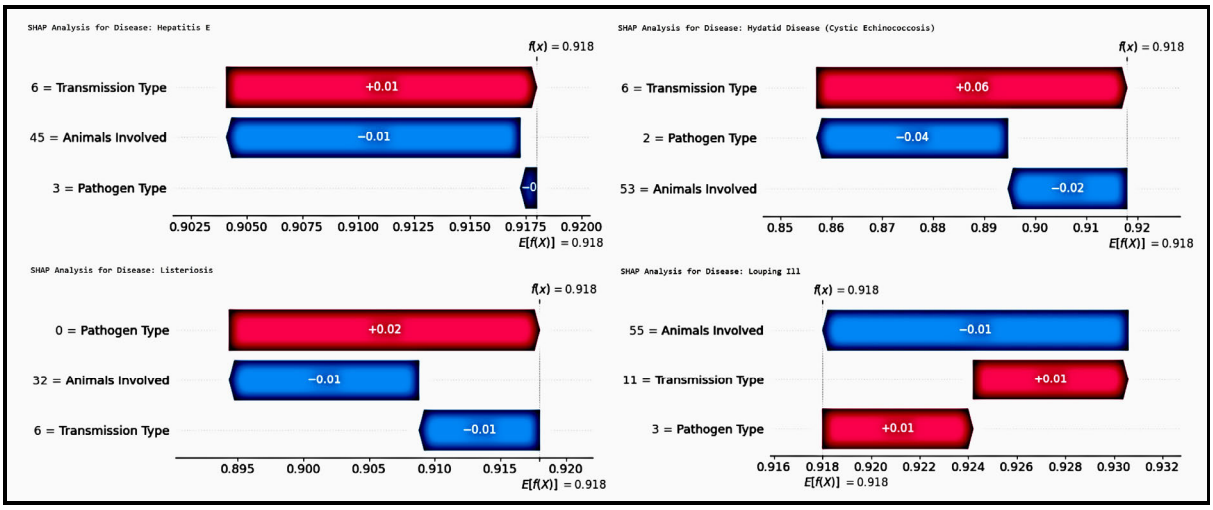


FIGURE 7. SHAP analysis of middle four diseases (Hepatitis E, Cystic Echinococcosis, Listeriosis, Louping Ill).

animal reservoirs are central determinants of zoonotic disease risk. In contrast,

Transmission Type consistently showed a negative effect on predictions, with SHAP values between −0.06 and −0.12. This suggests that certain transmission pathways either reduce the predicted risk level or that their influence is conditional on other epidemiological attributes, such as host species involvement. The Pathogen Type feature exhibited negligible impact across all four diseases, with SHAP values close to zero, indicating that broad pathogen classifications contribute minimally when transmission and host factors are considered.

Overall, the analysis demonstrates that while animal involvement remains the primary driver of zoonotic transmission risk, the magnitude of its effect varies depending on disease-specific ecological pathways. These findings emphasize the importance of targeted surveillance and management of high-risk animal reservoirs, rather than relying solely on pathogen categorization. Integrating ecological knowledge with epidemiological insights under a One Health framework will improve the identification of high-risk species and transmission routes, thereby enhancing strategies for zoonotic disease prevention and control.

1) COMPARATIVE INSIGHTS FROM SHAP ANALYSES

The three SHAP analyses collectively highlight how the influence of epidemiological factors varies across different zoonotic diseases. In the first set of diseases, Animals Involved emerged as the most dominant feature, reinforcing the critical role of host species in determining risk. In the second set,Transmission Type became the leading predictor, showing that certain transmission pathways can outweigh host-specific factors in shaping disease risk. In the third set, Animals Involved again dominated, with varying magnitudes of positive contribution across diseases, while Transmission Type consistently reduced predicted risk scores. Across all cases, Pathogen Type had minimal influence, with SHAP values close to zero, underscoring its limited predictive utility compared to host and transmission features.

These results demonstrate that while animal reservoirs are consistently central to zoonotic disease risk, the relative importance of transmission pathways shifts depending on disease context. This reinforces the need for disease-specific surveillance strategies, where host species monitoring is complemented by targeted investigation of transmission mechanisms. The findings further emphasize the importance of adopting a One Health approach, integrating ecological, epidemiological, and public health insights to achieve robust and context-sensitive risk assessments.

Figure 9 presents the ranked feature importance for rule-based zoonotic disease risk classification. The analysis highlights that *Contagious Period*, *Effective Antibiotics*, and *Diagnostic Test* are the most influential determinants of risk levels. A longer contagious period substantially increases transmission potential, thereby elevating overall risk classification. The availability or absence of effective antibiotics strongly modifies severity outcomes, while reliable diagnostic methods reduce uncertainty and enable timely containment. Features such as *Affected Age Group* and *Symptoms* also contribute significantly, emphasizing the role of host vulnerability and clinical presentation in risk determination. In contrast, broader descriptors such as

Pathogen Type and General Description exhibit minimal influence, underscoring that specific clinical, diagnostic, and epidemiological attributes drive the majority of rule-based risk categorization. This ranking validates the framework by aligning with established epidemiological reasoning, where transmissibility, treatment availability, and diagnostic capability are primary determinants of zoonotic disease risk.

In summary, the results demonstrate that while ensemble-based models such as Random Forest provide the most reliable predictive performance, SHAP analysis reveals that animal reservoirs and transmission pathways are the primary drivers of zoonotic disease risk. Complementing this, the

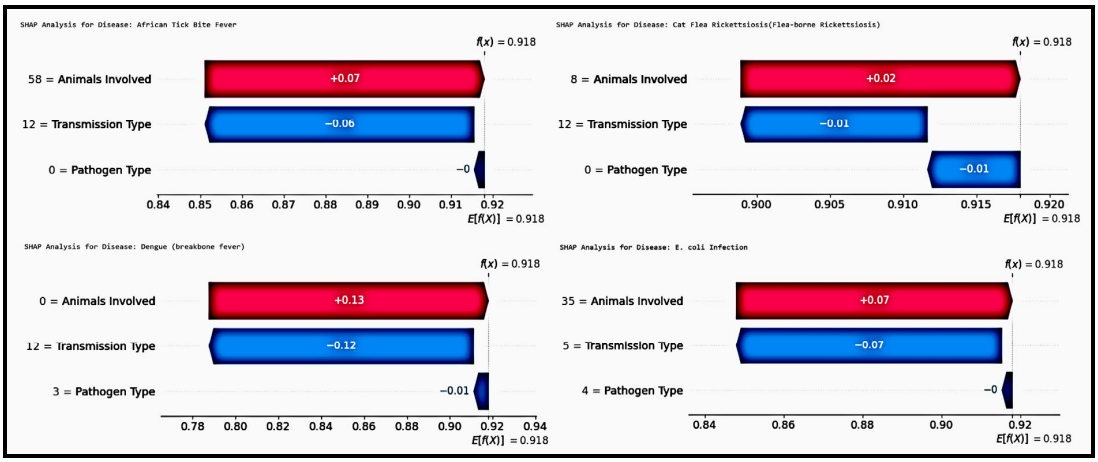


FIGURE 8. SHAP analysis of last four diseases (African Tick Byte Fever, Cat Flea Rickettsiosis, Dengue, E. coli).

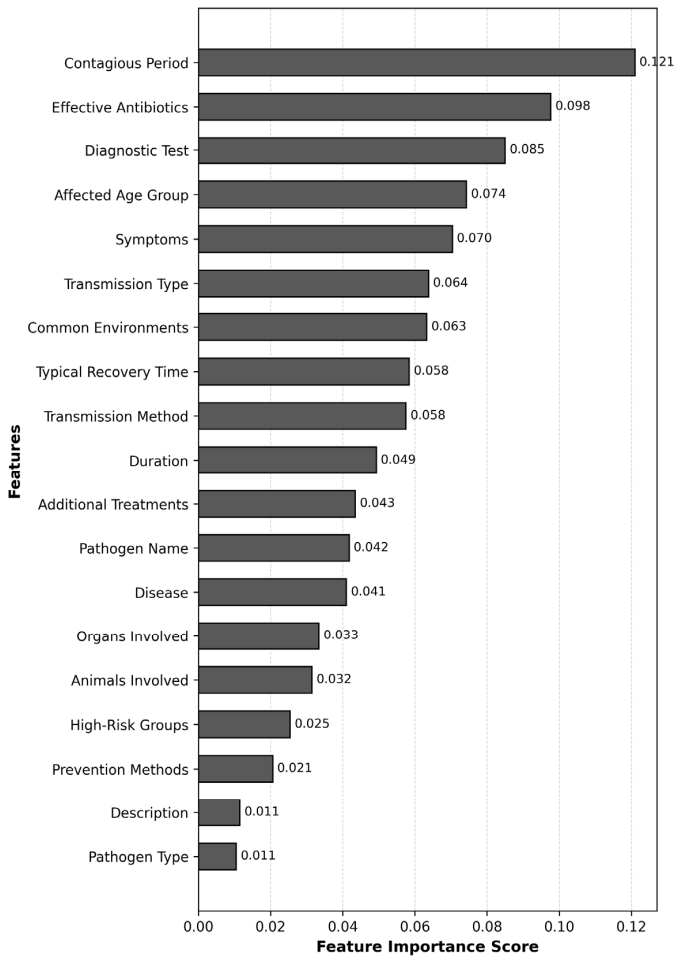


FIGURE 9. SHAP bar plot for feature importance ranking.

ranked feature importance (Figure. 9) highlights Contagious Period, Effective Antibiotics, and Diagnostic Test as the most influential determinants of risk classification. These findings align with epidemiological reasoning, where transmissibility, treatment availability, and diagnostic capability serve as critical determinants of outbreak potential and control. At the same time, the consistent underperformance in the Medium risk category underscores the challenge of class

imbalance, largely stemming from the underrepresentation of diseases within this category. Together, these results provide both methodological validation and epidemiological insight, reinforcing that predictive models must integrate host- and transmission-driven variables while also considering clinical and treatment-related factors. These outcomes form the basis for a broader discussion on improving predictive frameworks, addressing dataset limitations, and strengthening zoonotic disease surveillance through One Health–oriented strategies.

The top SHAP-ranked features—including contagious period, animals involved, diagnostic availability, and antibiotic effectiveness—are consistent with established epidemiological determinants of zoonotic transmission, underscoring the biological plausibility of the model outputs.

It is important to note that while the rule-based scoring framework emphasized five predefined dimensions (transmission type, animals involved, high-risk groups, contagious period, and symptom severity), the SHAP analysis highlighted a slightly different set of influential features, such as contagious period, availability of effective antibiotics, diagnostic confirmation, and animals involved. This variation reflects the complementary nature of the two approaches: the rule-based system is grounded in pre-defined scoring dimensions, whereas SHAP identifies feature contributions directly from data-driven model behaviour. The overlap in features such as contagious period and animals involved demonstrates consistency, while the divergence suggests that SHAP can reveal additional perspectives (e.g., treatment and diagnostic capacity) that enrich rather than contradict the framework.

D. RESULTS OF THE ZOONO-TRANSMIRISK TOOL

To validate the usability of the proposed framework, a prototype of the TransmiRisk tool was demonstrated using the compiled dataset of 63 zoonotic diseases with 21 features. Users interact with the tool via a Streamlit web interface, where a zoonotic disease can be selected from a dropdown menu or relevant epidemiological attributes can be entered directly. The system then generates a risk classification (Low, Medium, or High) using the integrated rule-based scoring system and compares it against predictions from machine learning models.

Risk assessment through the Zoono-TransmiRisk Framework is performed by evaluating parameters that influence zoonotic disease transmission, including transmission type, animal density, high-risk groups, and environmental conditions. Figures. 10–12 illustrate the tool's outputs through the Streamlit interface. Users provide these input parameters, which are then combined to generate a final risk score and corresponding categorical risk level.

The tool produces three distinct outputs based on input scenarios. Figure. 10 shows a low-risk case (Risk Score = 9) linked to food-borne transmission with low animal density. Figure. 11 presents a moderate-risk case (Risk Score = 10) associated with direct contact pathways. Figure. 12 demonstrates a high-risk case (Risk Score = 20) driven by airborne transmission and densely populated animal environments.

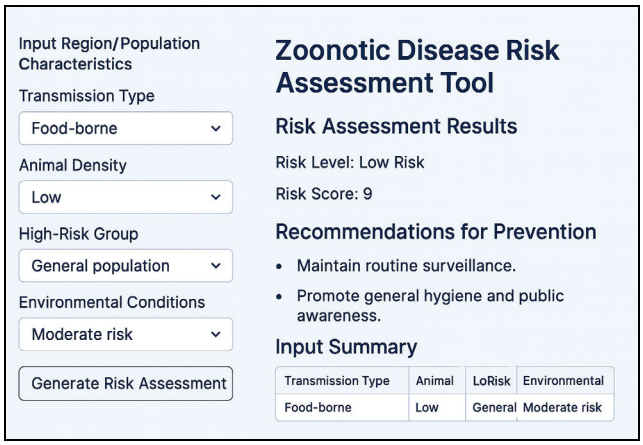


FIGURE 10. Zoono-TransmiRisk tool output for a low-risk scenario.

Each risk category is coupled with targeted recommendations: low-risk outputs emphasize routine surveillance and hygiene promotion, moderate-risk outputs suggest regional monitoring and resource preparedness, while high-risk outputs call for maximum resource allocation, vaccination programs, and education of high-risk populations. The interface also includes an input summary table, ensuring transparency in how risk levels are derived.

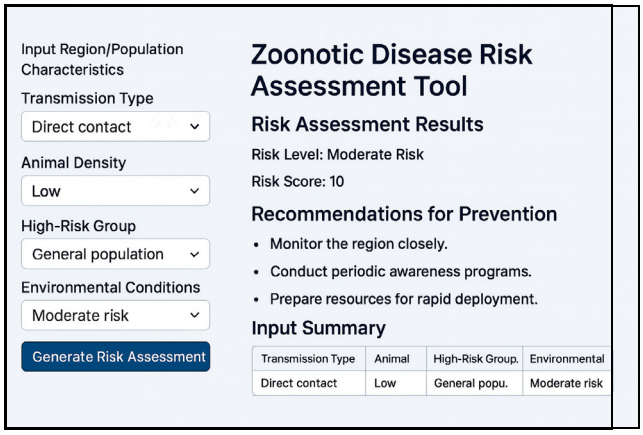


FIGURE 11. Zoono-TransmiRisk tool output for a moderate-risk scenario.

This demonstration highlights the practical utility of the tool for public health decision-making. By providing immediate evaluations with actionable recommendations, the Zoono-TransmiRisk Framework bridges technical risk modeling with field-level interventions. While effective in detecting scenarios of substantial transmission probability, its performance for moderate-risk classifications remains constrained by class imbalance, underscoring the need for further dataset expansion and methodological refinement.

V. DISCUSSION

Unlike traditional predictive studies focused on detecting the presence of specific diseases, the TransmiRisk tool is designed to classify transmission risk under user-defined conditions (e.g., transmission type, animal density, high-

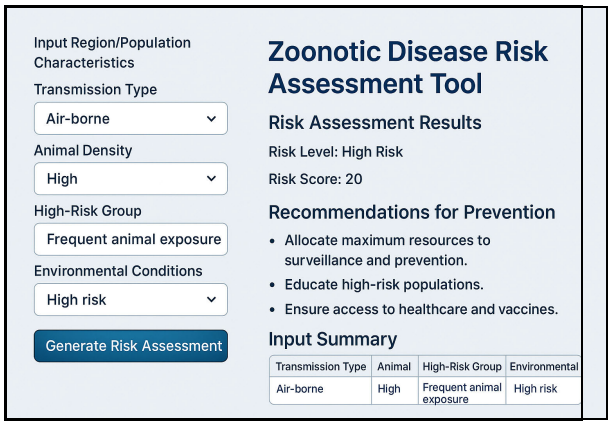


FIGURE 12. Zoono-TransmiRisk tool output for a high-risk scenario.

risk groups, environmental settings). This approach ensures practical alignment with public health needs, as the tool provides immediate, condition-based risk evaluations and recommends resource allocation strategies when higher transmission risks are detected. The findings of this study provide complementary insights from both the rule-based framework and the machine learning models. From the SHAP-based interpretability, the ranked feature importance (Figure. 9) identified Contagious Period, Effective Antibiotics, and Diagnostic Test as the most critical determinants of risk classification. These results align with established epidemiological reasoning, where transmissibility, treatment availability, and diagnostic capability directly shape outbreak potential and containment strategies. Features such as Affected Age Group and Symptoms also contributed notably, reinforcing the role of host vulnerability and clinical presentation in shaping disease risk. In contrast, broader descriptors such as Pathogen Type and General Description had minimal impact, underscoring that risk classification is more strongly driven by specific epidemiological and clinical attributes.

Some previous studies in zoonotic disease modelling have reported higher predictive performance, often approaching near-perfect accuracy. However, most of these works were designed around single-disease prediction tasks (e.g., influenza, brucellosis) with larger, patient-level datasets, which provide richer feature granularity and balanced class distributions. In contrast, the current study evaluates transmission risk classification across multiple zoonotic diseases using a disease-level dataset of 63 entries, which inherently introduces class imbalance (particularly the underrepresentation of Moderate and High categories). This broader generalization task is inherently more challenging, leading to comparatively lower performance metrics. Despite this, our framework contributes novelty by unifying rule-based scoring, machine learning validation, and SHAP-based interpretability into a single tool, offering insights beyond pure accuracy. The differences in results highlight that while disease-specific models may achieve higher predictive accuracy, they offer limited generalizability, whereas the TransmiRisk framework is oriented toward broader, condition-driven risk assessment across multiple zoonotic diseases.

From the machine learning perspective, Random Forest consistently outperformed XGBoost and Logistic Regression, achieving balanced precision, recall, and F1-scores. This demonstrates the robustness of ensemble-based methods in handling heterogeneous epidemiological datasets. Although XGBoost is designed to capture complex non-linear interactions, its performance was limited by the relatively small dataset size and class imbalance. Logistic Regression, while interpretable, struggled with the *Medium risk* class, confirming its limitations when applied to non-linear epidemiological processes. SHAP analysis further enhanced model interpretability by identifying Animals Involved and Transmission Type as dominant predictors across multiple disease groups, while Pathogen Type consistently exhibited minimal contribution.

Together, these results demonstrate that rule-based and machine learning approaches capture different but complementary aspects of zoonotic disease risk. The rule-based system emphasizes clinical, diagnostic, and treatment-related factors, while the machine learning models highlight host- and transmission-related drivers. This dual perspective strengthens the overall framework by ensuring that both epidemiological reasoning and data-driven learning contribute to risk characterization.

Despite these insights, limitations remain. The dataset exhibited a pronounced class imbalance, with only a few diseases categorized under *Medium* and *High* risk levels by the rule-based system. This underrepresentation directly contributed to the poor precision and recall observed in the *Medium* class across all models. Addressing this imbalance through oversampling, undersampling, or weighted loss functions will be critical for improving model performance. Furthermore, the reliance on categorical encoding limits the granularity of captured information, suggesting that future work should integrate richer data sources such as genomic, ecological, and environmental variables.

The comparison between the rule-based framework and SHAP outputs showed both convergence and complementarity: while both approaches emphasized contagious period and animals involved as critical drivers of zoonotic risk, SHAP additionally highlighted effective antibiotics and diagnostic availability, providing complementary insights that extend beyond the predefined scoring dimensions.

Finally, these findings reinforce the importance of a One Health approach to zoonotic disease surveillance. The dominance of host- and transmission-related features indicates the need for species-specific and context-sensitive monitoring, while the significance of diagnostic and treatment-related features highlights the role of health system capacity in shaping risk. Integrating ecological knowledge, clinical capacity, and epidemiological data into predictive frameworks will allow for more accurate and actionable assessments, ultimately strengthening public health interventions against emerging zoonotic threats.

The present study demonstrates a proof-of-concept for integrating rule-based scoring and machine learning in zoonotic risk prediction but does not yet scale to patient-level or real-time outbreak prediction. Future work will transition from disease-level to case-level surveillance data, integrate real-time feeds, and deploy the framework as a cloud-based Progressive Web App (PWA) for operational scalability.

A. LIMITATIONS

This study is subject to several limitations that should be considered when interpreting the results. A key limitation of this study is that the framework currently operates at the disease level, using structured epidemiological features rather than patient- or case-level data. While this enables systematic classification of transmission risks across multiple zoonotic diseases, it does not capture real-time individual-level variations. However, the scenario-driven design of the TransmiRisk tool ensures that users can still obtain actionable insights by inputting conditions such as transmission type, animal density, and high-risk groups, which better align with practical public health decision-making needs. The dataset may carry inherent biases due to geographic and socioeconomic underrepresentation, particularly for low-income regions. The dataset was imbalanced, with only one High risk instance. To enable stratified training and evaluation, this case was merged into the Moderate class. This limits the ability of the current framework to model High risk cases separately, highlighting the need for larger datasets with better class balance. Furthermore, potential class imbalance—particularly the underrepresentation of *Medium* and *High* risk categories—negatively affected model performance, as reflected in the poor detection of the *Medium* class by Logistic Regression and XGBoost.

The relatively small dataset size increased the risk of overfitting, even though stratified 5-fold cross-validation was employed. Since no external validation dataset was available, the evaluation relied solely on internal testing, which may limit the generalizability of findings. The dataset may also carry inherent biases due to the underrepresentation of specific geographic and socioeconomic contexts, particularly from low-income regions, which could restrict the broader applicability of the framework.

The current tool operates on static datasets and does not yet support real-time integration or automated updates. Similarly, although SHAP analysis was used for interpretability in this study, explainability modules are not yet embedded within the deployed prototype but are planned for future versions.

The study requires large-scale validation across diverse geographic regions and real-world public health settings to establish robustness and generalizability. Logistic Regression, while serving as a baseline, may oversimplify the non-linear relationships inherent in zoonotic disease transmission, further underscoring the need for more advanced approaches and richer datasets.

Another limitation of the present work is that deep learning models (e.g., Graph Neural Networks, LSTMs) could not be applied due to the small sample size and disease-level granularity of the dataset. Such methods require large, case level datasets to be effective and generalizable. Another limitation is that Logistic Regression oversimplifies complex zoonotic interactions due to its linear modelling assumptions, which contributed to reduced accuracy compared with non-linear ensemble models.

B. FUTURE WORK

The current version of the TransmiRisk framework has been implemented in Python with a Streamlit interface for research purposes. Future development will focus on converting the tool into a Progressive Web App (PWA), containerizing it with Docker, and deploying on cloud platforms such as AWS or Google Cloud. These improvements will enable scalability, integration with real-time data feeds, and eventual pilot deployment in health systems.

Future extensions will focus on integrating case-level surveillance data to improve predictive granularity and external validation. This will complement the current condition-based framework by enhancing real-time applicability for outbreak monitoring, thereby bridging the gap between scenario-driven transmission risk classification and patient-level epidemiological forecasting. The incorporation of deep learning approaches, particularly Graph Neural Networks (GNNs), will allow the modelling of complex zoonotic transmission networks beyond the current machine learning scope. Similarly, the application of advanced ensemble methods (e.g., Random Forest + XGBoost) and time-series models (e.g., LSTM) will further strengthen predictive accuracy and enable dynamic monitoring of outbreak trajectories.

To improve generalizability, the dataset will be expanded to encompass broader geographic regions and diverse socioeconomic contexts, thereby reducing bias and improving applicability in global health settings. Additionally, prioritizing the collection of larger datasets will enable the use of independent holdout sets or external validation cohorts, mitigating the risk of overfitting and strengthening model reliability. Future dataset expansion will prioritize inclusion of underrepresented geographic and socioeconomic contexts to reduce bias and enhance generalizability. Future extensions of the dataset will incorporate broader One Health drivers such as climate variability, land use change, and wildlife/bushmeat trade, enabling the framework to capture socio-environmental determinants of zoonotic spillover alongside core epidemiological attributes.

Finally, future extensions of the framework will continue to align with One Health principles, integrating epidemiological, clinical, and ecological perspectives. By combining risk characterization, machine learning, and interactive risk scoring, the TransmiRisk framework will evolve into a robust, scalable system for zoonotic disease surveillance and prevention.

VI. CONCLUSION

This study proposed an integrated framework for zoonotic disease risk assessment that combines a rule-based scoring system,machine learning models,and SHAP interpretability analysis. A rule-based scoring system was used to establish an interpretable baseline grounded in epidemiological principles, against which machine learning models were validated. The rule-based framework highlighted Contagious Period, Effective Antibiotics, and Diagnostic Test as the most influential determinants of risk classification, validating that transmissibility, treatment availability, and diagnostic capability are central to outbreak management. Additional factors such as Affected Age Group and Symptoms further emphasized the importance of host vulnerability and clinical presentation in risk assessment.

From the machine learning perspective, Random Forest achieved the most reliable performance, demonstrating balanced precision, recall, and F1-scores, while XGBoost and Logistic Regression were constrained by dataset size and class imbalance. SHAP analysis enhanced interpretability, revealing that *Animals Involved* and *Transmission Type* are dominant drivers of model predictions, whereas Pathogen Type consistently exerted minimal influence. These findings illustrate how rule-based and data-driven approaches capture complementary dimensions of zoonotic disease risk, thereby improving overall interpretability and robustness.

The study also identified critical limitations. The imbalance in risk categories, particularly the underrepresentation of *Medium* and *High* risk diseases, significantly hindered model performance. Addressing this issue through resampling techniques, weighted loss functions, and expansion of the dataset will be essential for improving predictive stability. Moreover, incorporating richer data sources such as genomic, ecological, and environmental attributes can further strengthen the framework and provide a deeper epidemiological basis for prediction. In conclusion, the proposed framework demonstrates that combining rule-based reasoning with machine learning and SHAP interpretability yields a powerful tool for zoonotic disease transmission risk assessment. By integrating epidemiological, clinical, and ecological dimensions within a One Health approach, this framework can support more accurate, interpretable, and actionable surveillance systems, thereby guiding public health preparedness and intervention strategies against emerging zoonotic threats.

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